

Project Sketch: HCI 5840

POTENTIAL PROJECT 1: Electric vehicle charging station locator. (second choice)

Describe the project in a few paragraphs:

The project is an electric vehicle (EV) Charging Station Locator designed to help electric vehicle drivers quickly find charging stations. The application would display stations on an interactive map and provide information such as distance, connector type, charging speed, cost, and availability. Users could filter stations based on their needs.

Who would be the users? Are there secondary stakeholders?

Primary Users:

- EV drivers who need to find charging stations.
- EV owners planning longer trips who want to identify charging stops ahead of time
- New EV drivers who may be unfamiliar with charging networks, connector types, or charging speeds

Secondary Stakeholders:

- Charging station operators.
- EV manufacturers
- Businesses that host charging stations
- Local government and transportation agencies
- Data/API providers

A major goal of the system would be to make finding a charging station less stressful, especially when a driver is running low on battery or needs to plan to travel long distances. Instead of having to check multiple charging-network apps or search through different websites, users could have the relevant information presented in one interface. Users can also filter results based on factors such as charging speed, connector type, price, distance, and station availability.

This would ideally be a mobile app. Charging station data could come from publicly available APIs or datasets, which would then process and display on an interactive map. The app could also potentially use their user's current location and vehicle preferences to prioritize useful stations.

Usage Story/Vignette:

A driver notices their EV battery is low and opens the application. They share their location, and the system displays nearby compatible charging stations. The driver filters for fast chargers and selects a station. The system shows the station's details and provides directions.

Problem:

The system helps reduce the difficulty of finding nearby, compatible, and available charging stations without having to search through multiple apps or websites.

Workflow:

Open app > enter location > view stations > apply filters > select station > view details/get directions. The user can repeat the search and adjust filters if the first results are suitable.

Data:

The system would use charging-station data such as location, connector type, charging speed, etc. the information could come from public API providers such as [NLR](#) or [Open Charge Map](#)